

Mechanics

L.O

Grade 1

Semester 1

Mechanics: Grade 1, Semester 1

Matter, Form and Function

Big Idea: The state of equilibrium of bodies requires a dynamic balance of forces.

ME.1.01 - Students will use position, displacement, average and instantaneous velocity, average and instantaneous acceleration to describe 1-dimensional motion of an object. (Week 01 - Week 03)

Essential Questions: ▶ What does it mean to stay still?

- ▶ Will it ever be the case that all observers agree on the motion of an object?
- ▶ What is the meaning of a negative velocity? And negative acceleration?

Skills:

- ▶ A. Use concepts of position, displacement, time, elapsed time, average velocity, instantaneous velocity, speed, and acceleration to understand and describe the movement of objects in 1D.
- ▶ B. Graphically determine average speed between two time points, estimate instantaneous velocity at one time point
- ▶ C. Compute relative velocities in classical mechanics (i.e., with speeds substantially less than the speed of light) in 1D
- ▶ D. Can explain the need for constant reference frame, and explain the signs of displacement, velocity, and acceleration within that frame.

Concepts:

- ▶ A. Position /Time graphs
- ▶ B. velocity/Time graphs
- ▶ C. Acceleration /Time graphs
- ▶ D. Relative velocity.
- ▶ E. Instantaneous velocity
- ▶ F. Average velocity
- ▶ G. Reference Frames

ME.1.02 - Students will use kinematic equations to understand and predict 1-dimensional motion of objects under constant acceleration, including vertical (free-fall) motion under gravity. (Week 04 - Week 08)

Essential Questions: Can one predict where an object will be at a future time if acceleration is not constant?

Skills:

- ▶ A. Explain why displacement equals area under a velocity-time graph.
- ▶ B. Use a velocity/time graph to explain, justify and/or derive kinematics equations for constant acceleration problems
- ▶ C. Solve 1-dimensional motion problems using the kinematic equations for constant acceleration.
- ▶ D. Solve kinematics problems involving vertical (free-fall) motion under gravity.

Concepts:

- ▶ A. Area under a curve
- ▶ B. Kinematic equations for 1-D motion with constant acceleration

- ▶ C. free-fall motion

ME.1.03 - Students will understand the importance of reference frames and use relative velocity to describe the motions of 2 objects with respect to each other in 1-D (Week 09 - Week 10)

Essential Questions: What happens to the relative velocity of a pool ball wrt a cue ball on an excellent pool shot that leaves the cue ball stationary?

Skills:

- ▶ A. Determine the relative velocity of one object wrt another, both of which moving in 1-D
- ▶ B. Can analyze the relative velocities of two objects before and after a collision in 1-D given the pre- and post-collision velocities

Concepts:

- ▶ A. Reference frame
- ▶ B. Reference object
- ▶ C. Velocity as a signed quantity
- ▶ D. Difference of velocities

ME.1.04 - Students will be able to use vector concepts to extend 1-d kinematics to motion in 2-D. (Week 11 - Week 14)

Essential Questions: Why do you study vectors?

Skills:

- ▶ A. Represent a vector using rectangular coordinates, polar coordinates, or engineering (i,j,k) notation
- ▶ B. Discriminate between vector quantities (e.g. displacement, velocity, force, acceleration) and scalar quantities that have magnitude but not direction (e.g. distance, speed)
- ▶ C. Translate fluently among different representations of a vector, including: rectangular coordinates, or engineering (i,j,k) notation, polar coordinates, physical diagrams, verbal descriptions, position vectors.
- ▶ D. Compute relative velocities in classical mechanics (i.e., with speeds substantially less than the speed of light) in 2D

Concepts:

- ▶ A. Vector and scalar quantities
- ▶ B. Rectangular coordinates
- ▶ C. Polar coordinates,
- ▶ D. Engineering (i,j,k) notation
- ▶ E. Physical diagrams
- ▶ F. Position vectors
- ▶ G. Resultant of two vectors
- ▶ H. Relative velocity in 2-dimensions